

A note on Problem #1 of the final exam.

If you interpret "a sequence of bounded functions ... in L^p " as $\exists M > 0$ s.t. $\|f_k\|_p \leq M \quad \forall k \in \mathbb{N}$, then there are two ways to prove the claim. Note that it suffices to show that $\int_{\Omega} f_k g \rightarrow \int_{\Omega} f g \quad \forall g \in L^q(\Omega)$. Also note that $f_k, f \in L^p(\Omega) \subset L^1(\Omega)$ since Ω is bounded in $\mathbb{R}^n$. Similarly, $g \in L^q(\Omega) \subset L^1(\Omega)$.

Pf #1. $\forall \varepsilon > 0$, $\exists \delta = \delta(\varepsilon) > 0$ and $K = K(\varepsilon, \delta) \in \mathbb{N}$, s.t.

(i). (by absolute continuity of the integral of $|g|^q$)

$$\left(\int_{E'} |g|^q \right)^{\frac{1}{q}} < \frac{\varepsilon}{4M} \quad \forall E' \subset \Omega \text{ satisfies } \mu(E') < \delta.$$

(ii) (by Egorov thm) $\exists E \subset \Omega$ satisfying $\mu(E) < \delta$ s.t.

$$f_k \rightrightarrows f \text{ in } \Omega \setminus E.$$

(iii) (by (ii)) $|f_k(x) - f(x)| < \frac{\varepsilon}{2\|g\|_1} \quad \forall k \geq K, x \in \Omega \setminus E.$

Now we take E' in (i) as E in (ii'), then

$$\left| \int_{\Omega} (f_k - f) g \right| \leq \int_{\Omega} |f_k - f| |g| = \int_{\Omega \setminus E} |f_k - f| |g| + \int_E |f_k - f| |g|$$

$$\leq \underbrace{\frac{\varepsilon}{2\|g\|_1} \int_{\Omega \setminus E} |g|}_{\text{by (iii)}} + \underbrace{\left(\int_E |f_k - f|^p \right)^{\frac{1}{p}} \left(\int_E |g|^q \right)^{\frac{1}{q}}}_{\text{by Hölder's inequality.}}$$

$$< \frac{\varepsilon}{2} + \underbrace{\|f_k - f\|_p}_{\leq \|f_k\|_p + \|f\|_p \leq 2M} \underbrace{\frac{\varepsilon}{4M}}_{\text{by (i)}} \leq \varepsilon$$

□

(WLOG we took $\|f\|_p \leq M$, otherwise just set $M = \max\{\|f_k\|_p, \|f\|_p\}$)

Pf #2. Since $\{f_k\} \subset \overline{B}(0; M) \subset L^p(\Omega)$ and $\overline{B}(0; M)$ is weakly closed, we know $\exists$ subseq $\{f_{k_j}\}$ of $\{f_k\}$ and $f_0 \in \overline{B}(0; M)$ s.t. $f_{k_j} \rightharpoonup f_0$.

Now we shall show that $f = f_0$ a.e. By Banach-Saks thm. $\exists$ subseq of $\{f_{k_j}\}$, still denoted by $\{f_{k_j}\}$ for notation simplicity, s.t. $\frac{1}{J} \sum_{j=1}^J f_{k_j} \rightarrow f_0$ in L^p as $J \rightarrow \infty$. Denote $g_J = \frac{1}{J} \sum_{j=1}^J f_{k_j}$. Then

$$g_J \rightarrow f_0 \text{ in } L^p \Rightarrow g_J \rightarrow f_0 \text{ in } L^1$$

$$\Rightarrow g_J \rightarrow f_0 \text{ in measure}$$

$$\Rightarrow \exists \text{ subseq } g_{J_k} \rightarrow f_0 \text{ pointwisely.}$$

On the other hand, $\forall x \in \Omega$, we know $f_{k_j}(x) \rightarrow f(x)$, and hence $g_{J_k}(x) = \frac{1}{J_k} \sum_{j=1}^{J_k} f_{k_j}(x) \rightarrow f(x)$ as $k \rightarrow \infty$

(Recall that if $\overset{\text{numerical seq}}{a_j} \rightarrow a$, then $\frac{1}{n} \sum_{j=1}^n a_j \xrightarrow{n \rightarrow \infty} a$).
Therefore $f_0 = f$ a.e.

We have showed that every weakly convergent subseq of $\{f_k\}$ has the weak limit f . It remains to show $f_k \rightharpoonup f$.

Suppose not, then $\exists F \in (L^p)^*$, $\varepsilon_0 > 0$, and a subseq $\{f_{k_j}\}$ s.t. $|F(f_{k_j}) - F(f)| \geq \varepsilon_0$. Since $\{f_{k_j}\} \subset \overline{B}(0; M)$,

it must have a weakly convergent subseq, still denoted by $\{f_{k_j}\}$, s.t. $f_{k_j} \rightharpoonup f$ as we proved above. This yields a contradiction. Hence $f_k \rightharpoonup f$. $\square$

Rmk. Note that the claim doesn't hold if each f_n has its own bound M_n i.e. $|f_n(x)| \leq M_n \quad \forall x \in \Omega, \forall n \in \mathbb{N}$.

A counter example is $f_k(x) = k \chi_{[0, \frac{1}{k}]}(x)$ defined on $\Omega = [0, 1]$.

Then $f_n \in L^2$, $\|f_k\|_2 = \sqrt{k} < \infty$, and $f_k \rightarrow 0$ a.e.

However, for $g(x) = x^{-\frac{1}{3}}$, we know $g \in L^2$, but

$$\int_0^1 f_n g = \int_0^{\frac{1}{k}} k x^{-\frac{1}{3}} dx = \frac{3}{2} k \left(x^{\frac{2}{3}} \Big|_0^{\frac{1}{k}} \right) \rightarrow \infty.$$